

Hints & Solutions

#MathBoleTohMathonGo

Q1 (3) - Single Correct

Since, $ax^2 + bx + 10 = 0$ has imaginary roots.

Therefore, $f(x) = ax^2 + bx + 10 \geq 0$ or ≤ 0 for all $x \in R$.

We have, $f(0) = 10 \geq 0$

$\therefore f(x) = ax^2 + bx + 10 \geq 0$ for all $x \in R$

$\Rightarrow f(5) = 25a + 5b + 10 \geq 0$

$\Rightarrow 5a + b + 2 \geq 0$

$\Rightarrow 5a + b \geq -2$

$\Rightarrow$ Minimum value of $5a + b$ is -2 .

$\Rightarrow 5\alpha + \beta = -2$

[$\because \alpha$ and β are the values of a and b for which $5a + b$ is minimum]

$\Rightarrow \beta = -5\alpha - 2$

Now, $\alpha(4x + 2y + 3) + \beta(x - y - 1) = 0$

$\Rightarrow \alpha(4x + 2y + 3) - (5\alpha + 2)(x - y - 1) = 0$

$\Rightarrow \alpha(-x + 7y + 8) - 2(x - y - 1) = 0$

Clearly, it represents a family of lines passing through the intersection of the lines $-x + 7y + 8 = 0$ and

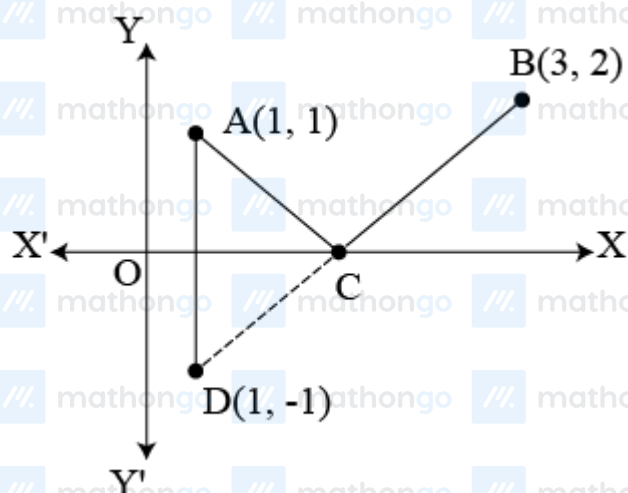
$x - y - 1 = 0$, i.e. the point $\left(-\frac{1}{6}, -\frac{7}{6}\right)$.

Q2 (1) - Single Correct

Clearly, $AC + BC$ will be minimum when points B , C and D (image of A in X -axis) are collinear. In that case, C is the intersection of BD with X -axis.

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The equation of BD is

$$y + 1 = \frac{2 + 1}{3 - 1}(x - 1)$$

$$\Rightarrow 3x - 2y - 5 = 0$$

It cut X -axis at $\left(\frac{5}{3}, 0\right)$. Hence, the coordinates of the required point are $\left(\frac{5}{3}, 0\right)$.

Q3 (1) - Single Correct

Let the coordinates of A be (h, k) . Then

$$d(A, 0) = 1$$

$$\Rightarrow \max\{|h|, |k|\} = 1$$

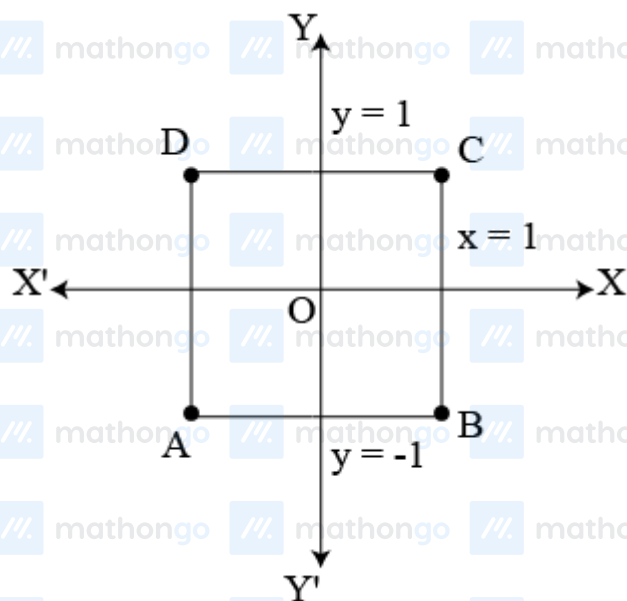
$$\Rightarrow |h| \leq 1 \text{ and } |k| \leq 1$$

$\Rightarrow$ Locus of $A(h, k)$ is $|x| \leq 1, |y| \leq 1$ Clearly, $|x| \leq 1$ and $|y| \leq 1$ represent the region enclosed by the

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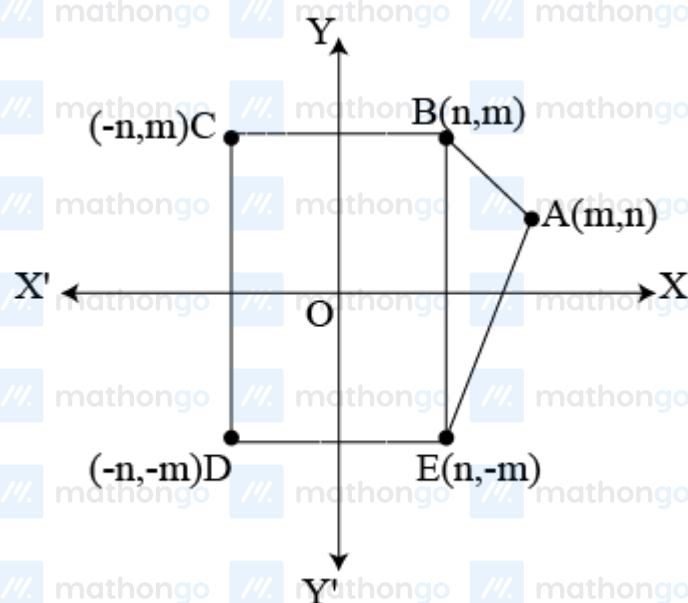
square shown in the figure given below:



Q4 (2) - Single Correct

Area of rectangle $BCDE$ is $4mn$.

$$\text{Area of } \triangle ABE = \frac{2m(m-n)}{2} = m^2 - mn$$



$$\therefore \text{Area of pentagon } ABCDE = 4mn + m^2 - mn$$

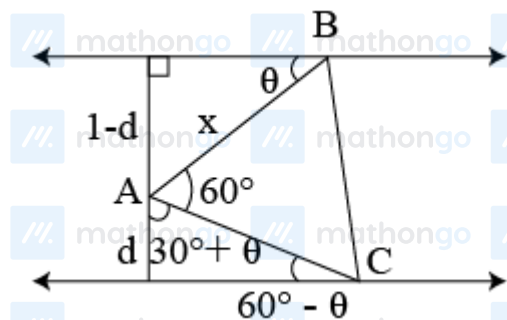
$$= m^2 + 3mn = m(m + 3n)$$

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Q5 (2) - Single Correct



From the figure,

$$x \cos(\theta + 30^\circ) = d \dots (i)$$

$$x \sin \theta = 1 - d \dots (ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\sqrt{3} \cot \theta = \frac{1+d}{1-d}$$

On squaring Eq.

(ii) and putting the value of $\cot \theta$, we have

$$x^2 = \frac{1}{3} (4d^2 - 4d + 4)$$

$$\Rightarrow x = 2\sqrt{\frac{d^2 - d + 1}{3}}$$

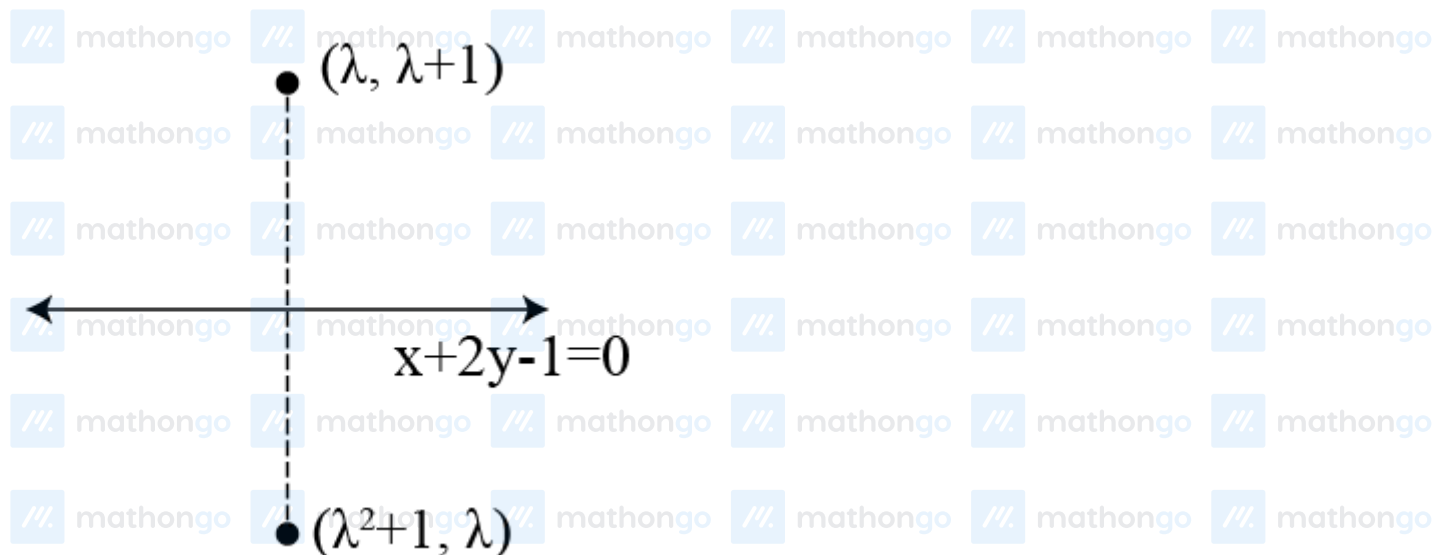
Q6 (1) - Single Correct

Since, $(\lambda^2 + 1, \lambda)$ is mirror image of $(\lambda, \lambda + 1)$, then

$$\frac{\lambda^2 + 1 - \lambda}{1} = \frac{\lambda - \lambda - 1}{2} = \frac{-2(\lambda + 2\lambda + 2 - 1)}{5}$$

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$$\Rightarrow \frac{\lambda^2 - \lambda + 1}{1} = -\frac{1}{2} = \frac{-2(3\lambda + 1)}{5}$$

$$\Rightarrow \frac{\lambda^2 - \lambda + 1}{5} = \frac{-6\lambda - 2}{5}$$

$$\text{and } \frac{-2(3\lambda + 1)}{5} = -\frac{1}{2}$$

$$\Rightarrow 5\lambda^2 - 5\lambda + 5 = -6\lambda - 2$$

$$\text{and } -6\lambda - 2 = -\frac{5}{2}$$

$$\Rightarrow 5\lambda^2 + \lambda + 7 = 0$$

$$\text{and } -6\lambda = -\frac{5}{2} + 2$$

$$\Rightarrow 5\lambda^2 + \lambda + 7 = 0$$

$$\text{and } \lambda = +\frac{5}{12} - \frac{1}{3}$$

$$\Rightarrow 5\lambda^2 + \lambda + 7 = 0 \text{ and } \lambda = \frac{1}{12}$$

Since, $\lambda = \frac{1}{12}$ does not satisfy $5\lambda^2 + \lambda + 7 = 0$

Hence, no value of λ is possible.

Q7 (4) - Single Correct

Equation of BC is $y = 2$, which is parallel to x -axis.

$$\therefore \tan \frac{B}{2} = \frac{4}{3} \Rightarrow B > \frac{\pi}{2}$$

$$\tan \frac{C}{2} = 2 \Rightarrow C > \frac{\pi}{2}$$

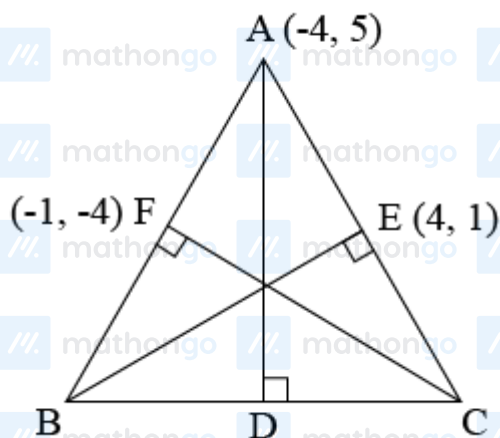
In a triangle, two angles cannot be greater than 90° and hence, there is no such triangle.

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Q8 (1) - Single Correct

$$\text{Slope of } AF = \frac{5 - (-4)}{-4 + 1} = -3$$



$$\text{Slope of } CF = \frac{1}{3}$$

Also, F lies on CF .

$$\therefore \text{Equation of } CF = x - 3y - 11 = 0 \dots(i)$$

$$\text{Slope of } AE = \frac{5-1}{-4-4} = -\frac{1}{2}$$

$$\text{Slope of } BE = \frac{-1}{-1/2} = 2. \text{ Also, } E \text{ lies on } BE.$$

$$\therefore \text{Equation of } BE = 2x - y - 7 = 0 \dots(ii)$$

On solving Eqs. (i) and (ii), we get point $O(2, -3)$. Since, CB is passing through intersection of AC and FC .

$$\text{Equation of } CB \Rightarrow (x + 2y - 6) + \lambda(x - 3y - 11) = 0$$

$$\text{Or } x(\lambda + 1) + y(2 - 3\lambda) - (6 + 11\lambda) = 0 \dots(iii)$$

$$\text{Slope of } CB = \frac{\lambda+1}{3\lambda-2}; \text{ also slope of } AO = \frac{5-(-3)}{-4-2} = -\frac{4}{3}$$

$$\text{Since, } CB \perp AO; \frac{\lambda+1}{3\lambda-2} \times -\frac{4}{3} = -1 \text{ or } \lambda = 2$$

$$\therefore \text{Equation of } BC \Rightarrow 3x - 4y - 28 = 0$$

Q9 (3) - Single Correct

$$\text{Area of } \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} m & n & 1 \\ 12 & 19 & 1 \\ 23 & 20 & 1 \end{vmatrix} = 70$$

$$\Rightarrow -m + 11n = 197 \pm 140 \dots(i)$$

Median through A has slope -5

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$$\Rightarrow 5m + n = 107 \dots(ii)$$

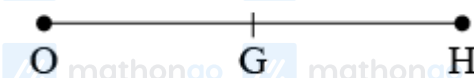
From Eqs. (i) and (ii), we get $m = 20, n = 7$

$$\Rightarrow m + n = 27$$

Q10 (1) - Single Correct

We know that, circumcentre, centroid and orthocentre of a triangle lie on a line, i.e. collinear and G divide OH in the ratio 1:2

i.e. $OG : GH = 1 : 2$



Let coordinates of H be (x, y)

By section formula, we have

Let coordinates of H be (x, y) By section formula, we have

$$\therefore 3 = \frac{x+12}{3}$$

$$\Rightarrow x + 12 = 9$$

$$\Rightarrow x = -3$$

$$\text{and } 3 = \frac{y+4}{3}$$

$$\Rightarrow y + 4 = 9$$

$$\Rightarrow y = 5$$

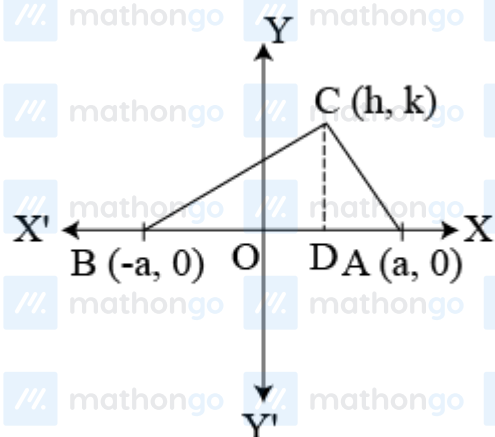
Hence, orthocentre is $(-3, 5)$.

Q11 (2) - Single Correct

We may suppose that, coordinates of two fixed points A and B are $(a, 0), (-a, 0)$ respectively and coordinates of variable point C are (h, k)

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Now, in $\triangle CDA$

$$\cot A = \frac{DA}{CD} = \frac{a-h}{k}$$

and in $\triangle DBC$,

$$\cot B = \frac{BD}{CD} = \frac{a+h}{k}$$

But $\cot A + \cot B = \text{Constant}$, C

$$\cot A + \cot B = \lambda \text{ (say)}$$

$$\frac{a-h}{k} + \frac{a+h}{k} = \lambda$$

$$\Rightarrow \frac{2a}{k} = \lambda$$

$$\Rightarrow 2a = k\lambda$$

$$\text{or } 2a = y\lambda$$

$$y = \frac{2a}{\lambda}$$

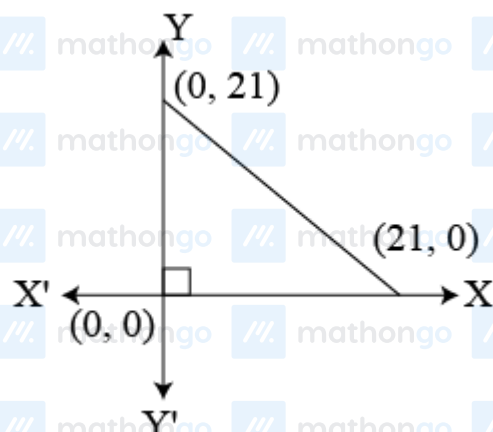
which is a line parallel to X -axis, i.e. AB .

Q12 (2) - Single Correct

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According to the figure,



where $x + y < 21$

$$0 < x < 21$$

$$0 < y < 21$$

$$\Rightarrow x + y + z = 21$$

where, $0 < x < 21$

$$0 < y < 21$$

$$0 < z < 21$$

$\therefore$ Total number of integral points

$$= \text{Coefficient } x^{21} \text{ in } (1 + x + x^2 + x^3 + \dots + x^{21})^3$$

$$= {}^{21-1}C_{3-1} = {}^{20}C_2 = \frac{20!}{2!18!} = \frac{20 \times 19}{2} = 190$$

Q13 (4) - Single Correct

We have, $y^3 - 4x^2y = 0$

$$\Rightarrow y(y^2 - 4x^2) = 0$$

$$\Rightarrow y(y + 2x)(y - 2x) = 0$$

$$\Rightarrow y = 0 \dots (i)$$

$$y + 2x = 0 \dots (ii)$$

$$y - 2x = 0 \dots (iii)$$

Since, these lines are concurrent.

$\therefore$ They do not form triangle.

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Q14 (4) - Single Correct

The equation of a line passing through the origin is

$$mx - y = 0 \dots(i)$$

If it is at a distance δ from (x_1, y_1) , then

$$\left| \frac{mx_1 - y_1}{\sqrt{m^2 - 1}} \right| = \delta$$

$$\Rightarrow (mx_1 - y_1)^2 = \delta^2 (m^2 - 1) \dots(ii)$$

$$\Rightarrow (yx_1 - xy_1)^2 = \delta^2 (x^2 + y^2) \quad \left(\because m = \frac{y}{x} \right)$$

Q15 (3) - Single Correct

We know that, the product of perpendiculars drawn from the point (x_1, y_1) to the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \dots(i)$$

$$\left| \frac{ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c}{\sqrt{(a-b)^2 + 4h^2}} \right|$$

$\therefore$ Product of perpendicular from $(0,0)$ to the lines (i) is

$$= \left| \frac{a \times 0 + 2h \times 0 \times 0 + b \times 0 + 2g \times 0 + 2f \times 0 + c}{\sqrt{(a-b)^2 + 4h^2}} \right|$$

$$= \left| \frac{c}{\sqrt{(a-b)^2 + 4h^2}} \right|$$

Q16 (1,3,4) - Multiple Correct

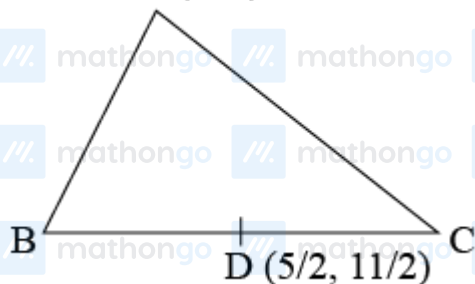
The coordinates of triangle are $A(1, -2), B(k, -2k)$

$C(1, 1 - 3)$

$$\frac{k+1+1}{3} = 2$$

$$\text{and } \frac{-2-2k+1-3}{3} = 3$$

$A(1, 2)$



$$\Rightarrow k = -3, l = 8$$

$\therefore B(-3, 6) \text{ and } C(8, 5)$

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Equation of BC is $x + 11y = 63$

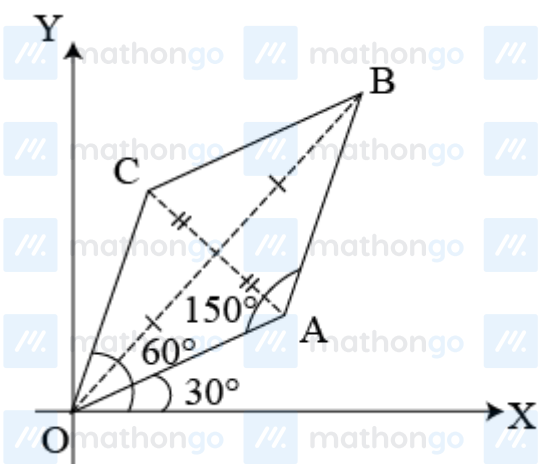
$$\Rightarrow p + q = 74$$

Medium through A , is $5x - y - 7 = 0$

Altitude through A , is $11x - y - 13 = 0$

Q17 (1,2) - Multiple Correct

Let the length of the sides of the rhombus be λ



$$\Rightarrow \text{Area} = \lambda^2 \cdot \sin 30^\circ = 2$$

$$\Rightarrow \lambda = 2$$

$$\therefore OA = OC = 2$$

$$\Rightarrow OB^2 = OA^2 + AB^2 - 2OA \cdot AB \cdot \cos 150^\circ$$

$$= 4 + 4 - 2(4) \cdot \left(-\frac{\sqrt{3}}{2}\right)$$

$$= 4(2 + \sqrt{3})$$

$$\therefore OB = 2\sqrt{2 + \sqrt{3}}$$

$$\Rightarrow B(\sqrt{4 + 2\sqrt{3}}, \sqrt{4 + 2\sqrt{3}})$$

$$= (1 + \sqrt{3}, 1 + \sqrt{3}) \in I \text{ quadrant}$$

$$B(-1 - 1\sqrt{3}, -1 - \sqrt{3}) \in III \text{ quadrant}$$

Q18 (1,2,3,4) - Multiple Correct

Given lines are $ax + by + c = 0$

$$bx + cy + a = 0$$

$$cx + ay + b = 0$$

Since, lines are concurrent.

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$$\therefore \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we get

$$\Rightarrow \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$, we get

$$\Rightarrow (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ b & c-b & a-b \\ c & a-c & b-c \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \{ (c-b)(b-c) - (a-b)(a-c) \} = 0$$

$$\Rightarrow (a+b+c) \{ (bc+ca+ab) - (a^2+b^2+c^2) \} = 0$$

$$\Rightarrow (a+b+c) [a^2+b^2+c^2 - ab - bc - ca] = 0$$

$$\Rightarrow a^2+b^2+c^2 - ab - bc - ca = 0 \quad (\because a+b+c \neq 0)$$

$$\Rightarrow a^2+b^2+c^2 = ab+bc+ca$$

$$\Rightarrow a=b=c$$

$$\text{Also, } (a+b+c)(a^2+b^2+c^2 - ab - bc - ca) = 0$$

$$\Rightarrow a^3+b^3+c^3 - 3abc = 0$$

Q19 (2,4) - Multiple Correct

The coordinates of vertices are

$$A\left(\frac{5}{4}, \frac{7}{8}\right), B(-7, 5) \text{ and } C\left(\frac{1}{3}, \frac{1}{9}\right)$$

$\therefore P(\alpha, \alpha^2)$ lies inside the $\triangle ABC$, then

(1) A and P must lie on the same side BC .

(2) B and P must lie on the same side of CA .

(3) C and P must lie on the same side of AB

Then,

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$$\frac{\frac{5}{2} + \frac{21}{8} - 1}{2\alpha + 3\alpha^2 - 1} > 0$$

$$\Rightarrow \frac{33}{2\alpha + 3\alpha^2 - 1} > 0$$

$$3\alpha^2 + 2\alpha - 1 > 0$$

$$(\alpha + 1) \left(\alpha - \frac{1}{3} \right) > 0$$

$$\Rightarrow \alpha \in (-\infty, -1) \cup \left(\frac{1}{3}, \infty \right) \dots (i)$$



$$\text{and } \frac{-35 - 30 - 1}{5\alpha - 6\alpha^2 - 1} > 0$$

$$\Rightarrow 5\alpha - 6\alpha^2 - 1 < 0$$

$$\Rightarrow \left(\alpha - \frac{1}{3} \right) \left(\alpha - \frac{1}{2} \right) > 0$$

$$\Rightarrow \alpha \in \left(-\infty, \frac{1}{3} \right) \cup \left(\frac{1}{2}, \infty \right) \dots (ii)$$



$$\text{and } \frac{\frac{1}{3} + \frac{2}{9} - 3}{\alpha + 2\alpha^2 - 3} > 0$$

$$\Rightarrow \alpha + 2\alpha^2 - 3 < 0$$

$$\Rightarrow (2\alpha + 3)(\alpha - 1) < 0$$

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$$\alpha \in \left(\frac{-3}{2}, 1 \right) \dots \text{(iii)}$$



From Eqs. (i), (ii) and (iii), we get

$$\alpha \in \left(\frac{-3}{2}, -1 \right) \cup \left(\frac{1}{2}, 1 \right)$$

Q20 (1,2) - Multiple Correct

Given family of lines, $(1 + 2\lambda)x - (1 + \lambda)y + 1 = 0$

$$\Rightarrow (x - y + 1) + \lambda(2x - y) = 0$$

$\therefore$ Family of lines passing through point of intersection of lines

$$x - y + 1 = 0 \text{ and } 2x - y = 0$$

i.e. through the point (1,2).

Since, the family of lines form an equilateral triangle with the line $x + y = 5$

$$\therefore y - 2 = \frac{-1 \pm \tan 60^\circ}{1 \mp (-1) \tan 60^\circ} (x - 1)$$

$$y - 2 = \frac{-1 \pm \sqrt{3}}{1 \mp (-\sqrt{3})} (x - 1) \Rightarrow y - 2 = \frac{-1 \pm \sqrt{3}}{1 \pm \sqrt{3}} (x - 1)$$

$$\text{and } y - 2 = \frac{-1 - \sqrt{3}}{1 - \sqrt{3}} (x - 1) \Rightarrow y - 2 = (2 - \sqrt{3})(x - 1)$$

$$\text{and } y - 2 = (2 + \sqrt{3})(x - 1)$$

Q21 (1,3) - Multiple Correct

Find the angular bisector of $x - \sqrt{3}y + 2\sqrt{3} = 0$ and $\sqrt{3}x - y + 3 = 0$. This will be parallel to the diagonal which is passing through (1,2) .

Q22 (1,3) - Multiple Correct

Let $C(h, k)$ be the third vertex of ΔABC which lies on the line $y = x + 3$

$$\therefore k = h + 3 \dots \text{(i)}$$

$$\text{Area of } \Delta ABC \Rightarrow \frac{1}{2} \begin{vmatrix} 2 & 1 & 1 \\ 3 & -2 & 1 \\ h & k & 1 \end{vmatrix} = \pm 5$$

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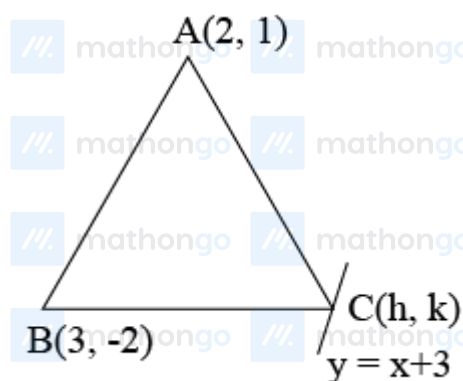
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$$\Rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 3 & -2 & 1 \end{vmatrix} = \pm 10$$

$$\Rightarrow \begin{vmatrix} h & k & 1 \\ 2 & 1 & 1 \\ 1 & -3 & 0 \end{vmatrix} = \pm 10$$

$$\Rightarrow \begin{vmatrix} h & k & 1 \\ h-2 & k-1 & 0 \end{vmatrix}$$



Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Rightarrow (k-1) + 3(h-2) = \pm 10$$

$$\Rightarrow k - 1 + 3h - 6 = \pm 10$$

$$\Rightarrow 3h + k - 7 = \pm 10$$

$$\Rightarrow 3h + k - 7 = 10$$

$$\text{or } 3h + k - 7 = -10$$

$$\Rightarrow 3h + k = 17 \text{ or } 3h + k = -3$$

$$\Rightarrow 3h + k = 17 \dots \text{(ii)}$$

$$\text{and } 3h + k = -3 \dots \text{(iii)}$$

On solving Eqs. (i) and (ii), we get

$$h = \frac{7}{2}, k = \frac{13}{2}$$

On solving Eqs. (i) and (iii), we get

$$h = -\frac{3}{2}, k = \frac{3}{2}$$

Hence, coordinates of third vertex are $\left(\frac{7}{2}, \frac{13}{2}\right)$ and $\left(-\frac{3}{2}, \frac{3}{2}\right)$

Q23 (1,3) - Multiple Correct

Given, equation of lines are

$$x + y = 0 \dots \text{(i)}$$

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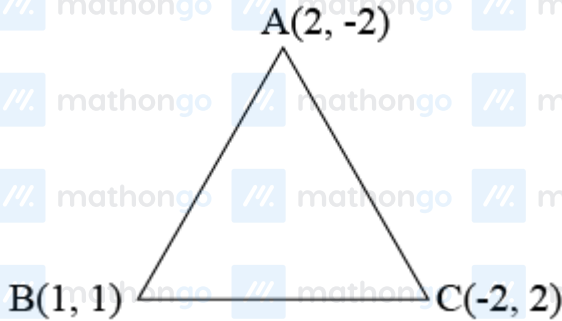
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$$3x + y - 4 = 0 \dots(ii)$$

$$\text{and } x + 3y - 4 = 0 \dots(iii)$$

Taking mutual intersection of lines (i), (ii) and (iii), we get the vertices of triangles as (2,-2), (1,1) and (-2,2)



Now,

$$AB = \sqrt{(2-1)^2 + (-2-1)^2} = \sqrt{10}$$

$$BC = \sqrt{(3)^2 + (-1)^2} = \sqrt{10}$$

$$AC = \sqrt{(2+2)^2 + (-2-2)^2} = \sqrt{32}$$

$$AB = BC$$

$\therefore \Rightarrow \Delta ABC$ is isosceles

Hence, option (1) is correct.

Now,

$$\cos B = \frac{AB^2 + BC^2 - AC^2}{2AB \cdot BC}$$

$$= \frac{10+10-32}{2 \times 10} = \frac{-12}{20} = -\frac{3}{5}$$

$$\cos B = -\frac{3}{5}$$

$$\angle B > 90^\circ \left[\because \cos \theta < 0, \text{ in } \theta \in \left(\frac{\pi}{2}, \frac{3\pi}{2} \right) \right]$$

$\Rightarrow \Delta ABC$ is obtuse angled triangle.

Hence, option (3) is correct.

Q24 (1,2) - Multiple Correct

$$\text{Given, } x^2 - 3|x| + 2 = 0 \dots(i)$$

$$y^2 - 3y + 2 = 0 \dots(ii)$$

From Eq. (i), we have

$$|x|^2 - 3|x| + 2 = 0$$

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$$\Rightarrow |x| = 2, 1$$

$$|f(x)| = 2$$

$$\Rightarrow x = \pm 2$$

$$\text{and } |x| = 1$$

$$\Rightarrow x = \pm 1$$

From Eq. (ii), we have

$$y^2 - 3y + 2 = 0$$

$$\Rightarrow y = 2, 1$$

$\therefore$ Coordinates of vertices of square are

$$(2, 2), (2, 1), (-2, 2), (-2, 1), (1, 2), (1, 1), (-1, 2), (-1, 1)$$

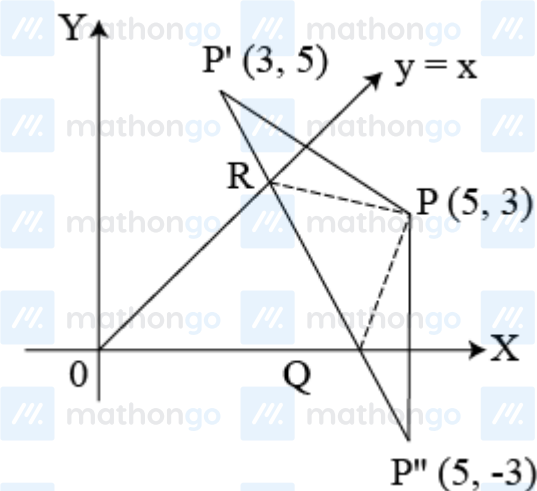
Hence, options (1) and (2) are correct.

Q25 (1) - Multiple Correct

P' , P'' are images of P w.r.t. $y = x$, $y = 0$, respectively.

$$\therefore P'(3, 5) \text{ and } P''(5, -3)$$

Equation of $P'P''$, is



$$y + 3 = \frac{5 + 3}{5 - 3}(x - 5)$$

$$\therefore Q = 5 + 3 \left(-\frac{1}{4} \right) = \frac{17}{4}$$

$$\therefore Q = \left(\frac{17}{4}, 0 \right)$$

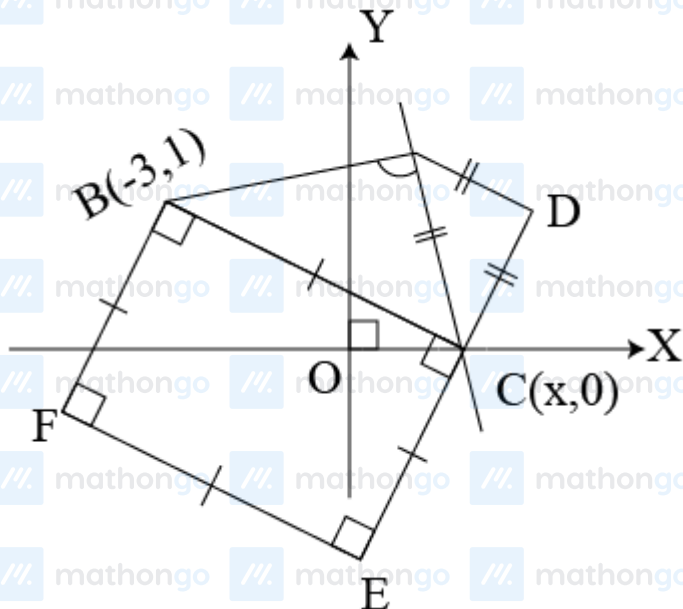
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Q26 (4) - Paragraph

Explanation (For Question 26, 27)

Since, $A = 90^\circ$



$$\therefore m_{AC} \cdot m_{AB} = -1$$

$$\Rightarrow \frac{2}{1-x} \times \frac{1}{4} = -1$$

$$\Rightarrow x = \frac{3}{2}$$

$$\Rightarrow C\left(\frac{3}{2}, 0\right)$$

Also, by rotation

$$F\left(-4, \frac{-7}{2}\right), E\left(\frac{1}{2}, \frac{-9}{2}\right) \text{ and } D\left(\frac{5+4\sqrt{3}}{4}, \frac{4+\sqrt{3}}{4}\right)$$

$$\text{Slope of } AF = \frac{2+\frac{7}{2}}{1+4} = \frac{11}{10}$$

Q27 (3) - Paragraph

Abcissa of centroid of $\triangle BCE$,

$$= \left(\frac{-3+\frac{3}{2}+\frac{1}{2}}{3}\right) = \frac{-1}{3}$$

Q28 (4) - Paragraph

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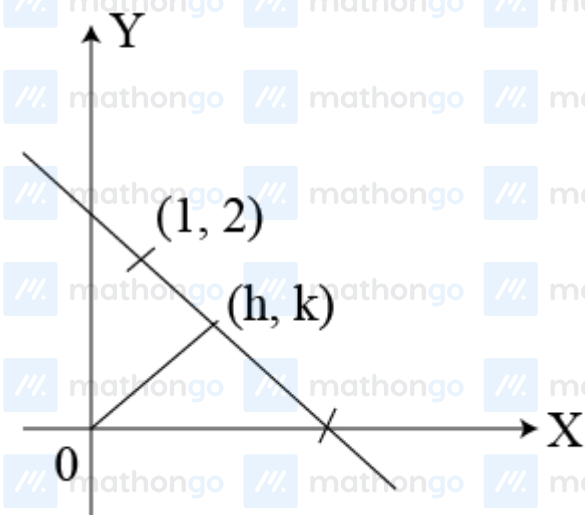
Explanation (For Question 28, 29)

$$\text{Given, } (4a + 3)x - (a + 1)y - (2a + 1) = 0$$

$$\Rightarrow (3x - y - 1) + a(4x - y - 2) = 0$$

Family of lines passing through the fixed point $P(1, 2)$

$$\frac{k}{h} \cdot \left(\frac{k-2}{h-1} \right) = -1$$



$$\therefore \text{Locus is } x(x - 1) + y(y + 2) = 0$$

$$\Rightarrow x^2 + y^2 - 2y - x = 0$$

$$\Rightarrow \left(x - \frac{1}{2} \right)^2 + (y - 1)^2 = \frac{5}{4}$$

$$\text{Or } (2x - 1)^2 + 4(y - 1)^2 = 5$$

Q29 (1) - Paragraph

$$\text{We have, } (y - 2) = m(x - 1)$$

This makes an angle of $\frac{\pi}{4}$ with $3x - 4y = 2$ with slope $3/4$.

$$\therefore \left| \frac{m - 3/4}{1 + \frac{3}{4}m} \right| = 1$$

$$\Rightarrow m = 7, -\frac{1}{7} \text{ (neglecting } -1/7)$$

$$\text{Hence, the line is } y - 2 = 7(x - 1)$$

$$\Rightarrow 7x - y - 5 = 0$$

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Q30 (4) - Paragraph

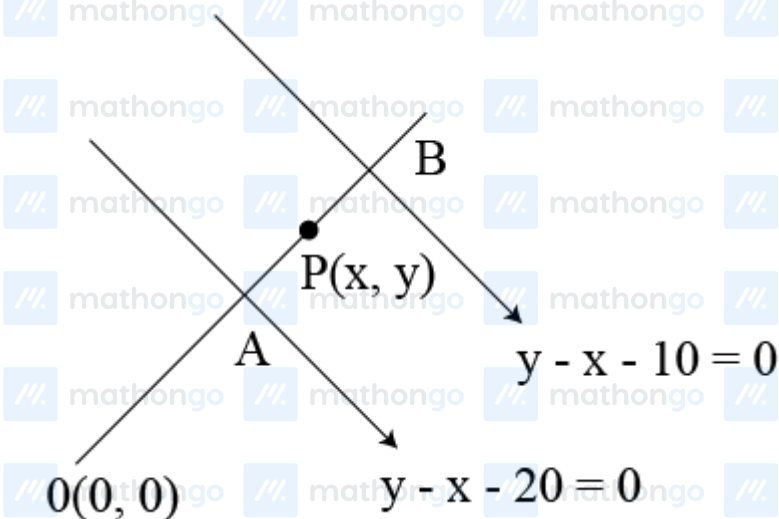
Let $OA = r_1, OP = r, OB = r_2$

$A(r_1 \cos \theta, r_1 \sin \theta)$

$B(r_2 \cos \theta, r_2 \sin \theta)$

$P(x, y) = (r \cos \theta, r \sin \theta)$

A lie on $y - x - 20 = 0$



$$\therefore r_1 \sin \theta - r_1 \cos \theta - 20 = 0$$

$$\frac{20}{r_1} = \sin \theta - \cos \theta$$

$$\text{Also, } \frac{10}{r_2} = \sin \theta - \cos \theta$$

$$\therefore \frac{2}{OP} = \frac{1}{OA} + \frac{1}{OB}$$

$$\Rightarrow \frac{2}{r} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{3 \sin \theta - 3 \cos \theta}{20}$$

$$\therefore 3(r \sin \theta) - 3(r \cos \theta) = 40 \Rightarrow 3y - 3x = 40$$

Q31 (3) - Paragraph

$$OP^2 = OA \cdot OB$$

$$r^2 = \frac{10 \times 20}{(\sin \theta - \cos \theta)^2}$$

$$\Rightarrow (r \sin \theta - r \cos \theta)^2 = 200$$

Hence, locus is $(y - x)^2 = 200$

Q32 (4) - Paragraph

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$$\therefore OR = AR$$

$$\Rightarrow |x - 0| + |y - 0| = |x - 1| + |y - 2|$$

$$\Rightarrow |x| + |y| = |x - 1| + |y - 2|$$

$$\text{as } 0 \leq x < 1, 0 \leq y < 2$$

$$\therefore x + y = 1 - x + 2 - y$$

$$\Rightarrow 2x + 2y = 3$$

Q33 (4) - Paragraph

$$\therefore OS = OR$$

$$\Rightarrow |x| + |y| = |x - 2| + |y - 3|$$

$$\therefore x \geq 2 \text{ and } 0 \leq y < 3$$

$$\Rightarrow x + y = x - 2 + 3 - y$$

$$\Rightarrow y = \frac{1}{2}$$

Q34 (2) - Paragraph

$$\text{Given, } f(\alpha, \beta) \leq 0 \dots (i)$$

$$\text{Also, } f(\alpha, \beta) = \left| \frac{\beta}{\alpha} - \frac{3}{2} \right| + (3\alpha - 2\beta)^6 + \sqrt{e\alpha + 2\beta - 2e - 6} \geq 0 \dots (ii)$$

$$\text{From Eqs. (i) and (ii), } f(\alpha, \beta) = 0$$

$$\Rightarrow \frac{\beta}{\alpha} = \frac{3}{2}, 3\alpha - 2\beta = 0$$

$$\text{and } e\alpha + 2\beta - 2e - 6 = 0$$

$$\text{On solving, } 2\beta = 3\alpha \text{ and } e\alpha + 2\beta - 2e - 6 = 0, \text{ we get } \alpha = 2, \beta = 3$$

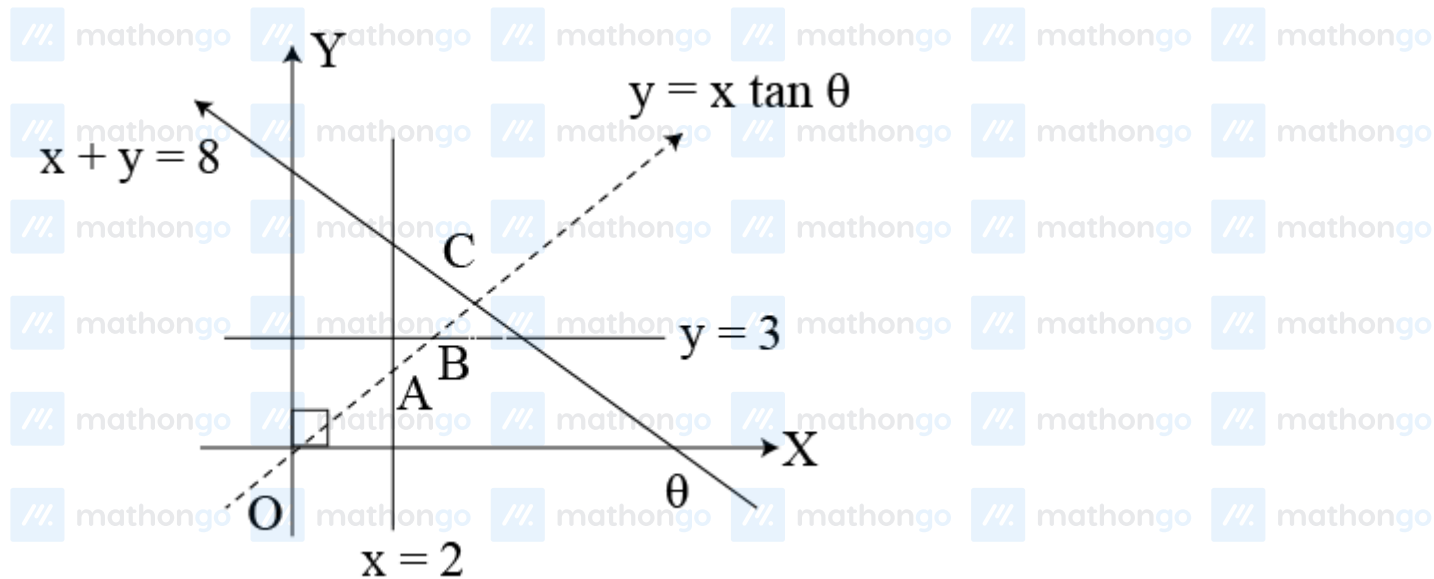
Q35 (1) - Paragraph

$$\text{Given, } x = \alpha, y = \beta \text{ and } x + y = 8$$

$$\text{i.e. } x = 2, y = 3 \text{ and } x + y = 8$$

Hints & Solutions

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Let $OA = r_1$, $OB = r_2$ and $OC = r_3$

$\therefore A(r_1 \cos \theta, r_1 \sin \theta)$, $B(r_2 \cos \theta, r_2 \sin \theta)$, $C(r_3 \cos \theta, r_3 \sin \theta)$

$\Rightarrow r_1 \cos \theta = 2$, $r_2 \sin \theta = 3$ and $r_3(\cos \theta + \sin \theta) = 8$

Given, $r_1 r_2 r_3 = 48\sqrt{2}$

$\Rightarrow \frac{2}{\cos \theta} \cdot \frac{3}{\sin \theta} \cdot \frac{8}{(\cos \theta + \sin \theta)} = 48\sqrt{2}$

$\Rightarrow \sqrt{2} = \sin 2\theta \cdot (\sin \theta + \cos \theta)$

which is only possible, when $\theta = 45^\circ$. $\therefore y = x$ is required straight line.

Q36 (3) - Paragraph

Since, diagonal is member of both the families.

$\therefore$ it passes through $(1,0)$ and $(-1,1)$

Hence, equation of diagonal AC is $x + 2y - 1 = 0$

Q37 (2) - Paragraph

$\therefore (3, 2)$ does not lie on AC .

$\therefore$ It lies on BD .

Hence, equation of BD is $2x - y = 4$.

Q38 (0) - Integer Type

Hints & Solutions

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The equation of line, which passes through (4, -5) having slope m , is

$$y + 5 = m(x - 4)$$

$$\Rightarrow y + 5 = mx - 4m$$

$$\Rightarrow mx - y - 4m - 5 = 0 \quad \dots (i)$$

It is given that the distance of point (-2,3) from line (i) is 12 units. $\therefore \frac{|-2m-3-4m-5|}{\sqrt{m^2+1}} = 12$

$$\Rightarrow \frac{|-6m-8|}{\sqrt{m^2+1}} = 12$$

$$\Rightarrow \frac{-6m-8}{\sqrt{m^2+1}} = \pm 12$$

Taking positive sign, we get

$$-6m - 8 = 12(\sqrt{m^2+1})$$

$$(6m+8)^2 = 144(m^2+1)$$

$$\Rightarrow 36m^2 + 64 + 96m = 144m^2 + 144$$

$$\Rightarrow 108m^2 - 96m + 80 = 0$$

$$\Rightarrow 27m^2 - 24m + 20 = 0$$

$\Rightarrow$ Since, $D < 0$. Therefore real value of m is not possible.

Hence, no lines are possible.

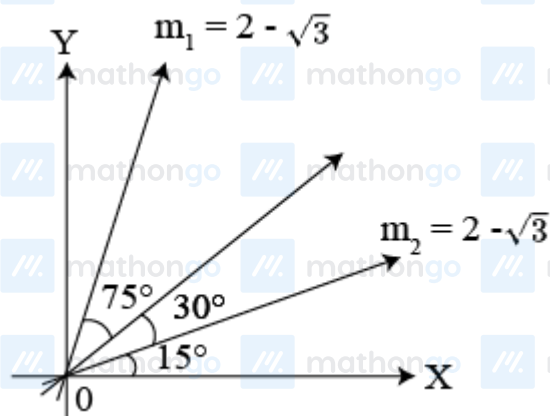
Q39 (6) - Integer Type

Given, $x^2 - 4x + 1 = 0$

$$\Rightarrow x = 2 + \sqrt{3}$$

From below figure, $l(a, b)$ lies on line $y = x$. So, $b = a$.

Also, from $\triangle IBD$.



$$\frac{r}{BI} = \sin 30^\circ$$

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Hints & Solutions

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$$\Rightarrow BI = 2 \quad (\because \text{as } r = 1)$$

$$\therefore BI = 2 = \sqrt{a^2 + a^2}$$

$$\Rightarrow a = \sqrt{2}$$

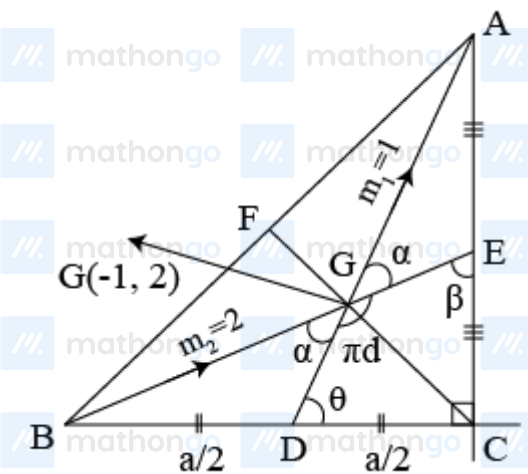
$$\text{So, } I(\sqrt{2}, \sqrt{2}) = (a, b)$$

$$\text{Hence, } a^2 + b^4 = 6$$

Q40 (4) - Integer Type

$$\text{Area} = \frac{1}{2}ab$$

$$AD : y = x + 3, \quad BE : y = 2x + 4$$



$$\text{Solving } G(-1, 2)$$

Acute angle α between the medians is

$$\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\Rightarrow \tan \alpha = \frac{2-1}{1+2} = \frac{1}{3}$$

$$\text{Now, } (180 - \alpha) + 90^\circ + \theta + \beta = 360^\circ$$

$$\Rightarrow \alpha = \theta + \beta - 90^\circ$$

$$\cot \alpha = -\tan(\theta + \beta)$$

$$-3 = \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta}$$

$$\Rightarrow -3 = \frac{\frac{2b}{a} + \frac{2a}{b}}{1 - \frac{2b}{a} \cdot \frac{2a}{b}}$$

$$\Rightarrow 9 = \frac{2(a^2 + b^2)}{ab}$$

$$\Rightarrow 9ab = 2(3600)$$

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$$\Rightarrow \frac{ab}{2} = 400 = abc$$

$$\therefore a + b + c = 4$$

Q41 (1) - Integer Type

$$P\left(\frac{\alpha \cos \alpha - \sin \alpha}{\sin \alpha - \alpha}, \frac{\sin^2 \alpha - \alpha^2 \cos \alpha}{(\sin \alpha - \alpha) \cdot \sin \alpha \cos \alpha}\right)$$

As limit, $\alpha \rightarrow 0$

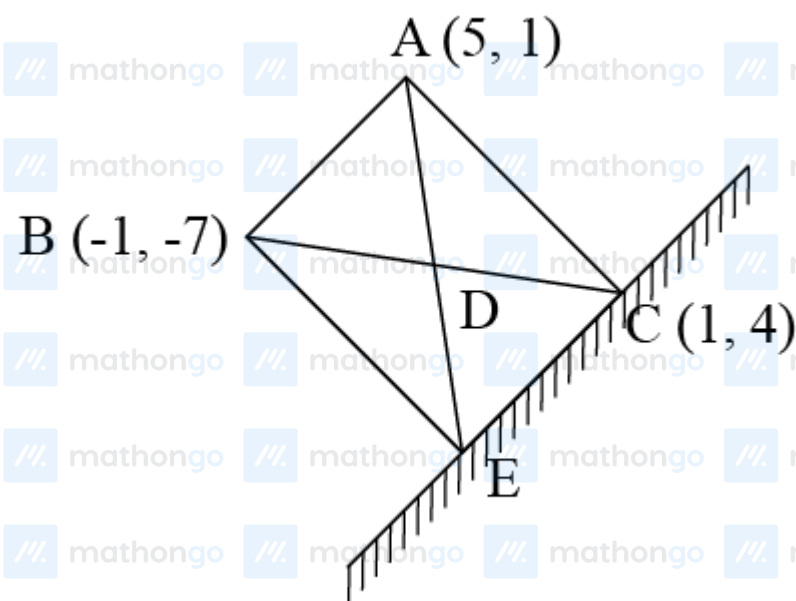
$$P(2, -1)$$

$$\therefore (h + k) = 1$$

Q42 (9) - Integer Type

$$\frac{AD}{DE} = \frac{BD}{DC} = \frac{AB}{CE}$$

$$\therefore \text{Point } D\left(\frac{1}{3}, \frac{1}{3}\right) \text{ and } E(-2, 0).$$



$$\text{Area of } \triangle ACE + \text{area } \triangle ABC = 2 \text{ area } \triangle ACE$$

$$= 2 \times \frac{19}{2} = 19 = 10 + 9$$

$$\therefore a = 9$$

Q43 (8) - Integer Type

The line $y = \sqrt{3}x$ can be written as $x = r/2, y = \frac{r\sqrt{3}}{2}$. If this line cuts the given curve, then

$$\frac{r^4}{16} + \frac{ar^3\sqrt{3}}{8} + \frac{br^2\sqrt{3}}{4} + \frac{cr}{2} + \frac{dr\sqrt{3}}{2} + 6 = 0$$

$$\therefore \frac{OA \cdot OB \cdot OC \cdot OD}{12} = \frac{1}{12} \cdot |r_1 r_2 r_3 r_4| = 8$$

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Hints & Solutions

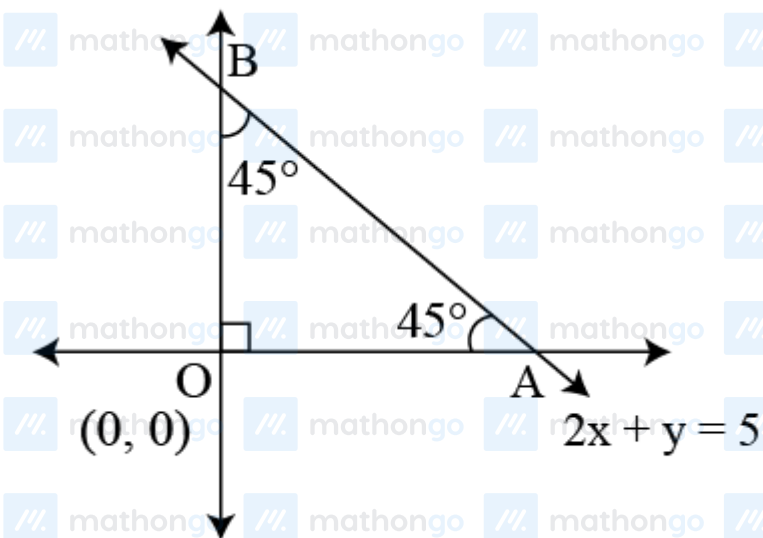
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Q44 (5) - Integer Type

Given, equation of line

$$2x + y = 5 \dots (i)$$

Slope of line (i), $m = -2$



$\therefore$ Equation of lines passing through $(0,0)$ and making an angle 45° with the line AB , is

$$y - 0 = \frac{-2 \pm \tan 45^\circ}{1 \mp (-2) \tan 45^\circ} (x - 0)$$

$$\Rightarrow y = \frac{-2 \pm 1}{1 \mp (-2)} x \Rightarrow y = \frac{-2 + 1}{1 - (-2)} x$$

$$\text{and } y = \frac{-2 - 1}{1 + (-2)} x$$

$$\Rightarrow y = -\frac{1}{3}x \text{ and } y = -\frac{3}{1}x$$

$$x + 3y = 0 \text{ and } 3x + y = 0$$

Now, to get the coordinates of point A , on solving equation $2x + y = 5$ and $x + 3y = 0$, we get

$$x = 3, y = -1$$

$\therefore$ Coordinates of point A are $(3, -1)$.

$$\text{Now, } OA = \sqrt{(3 - 0)^2 + (-1 - 0)^2} \\ = \sqrt{9 + 1} = \sqrt{10}$$

Since, $\triangle OAB$ is isosceles triangle, right angled at O .

$$\therefore OA = OB = \sqrt{10} \text{ and } \angle AOB = 90^\circ$$

Hints & Solutions

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Hence, area of $\Delta AOB = \frac{1}{2} \times OA \times OB$

$$= \frac{1}{2} \times \sqrt{10} \times \sqrt{10} = 5 \text{ sq units}$$

Q45 (5) - Integer Type

Since, $\text{ar}(\Delta PAB) = \text{ar}(\Delta PBC) = \text{ar}(\Delta PCA)$

$\therefore P$ must be centroid of ΔABC

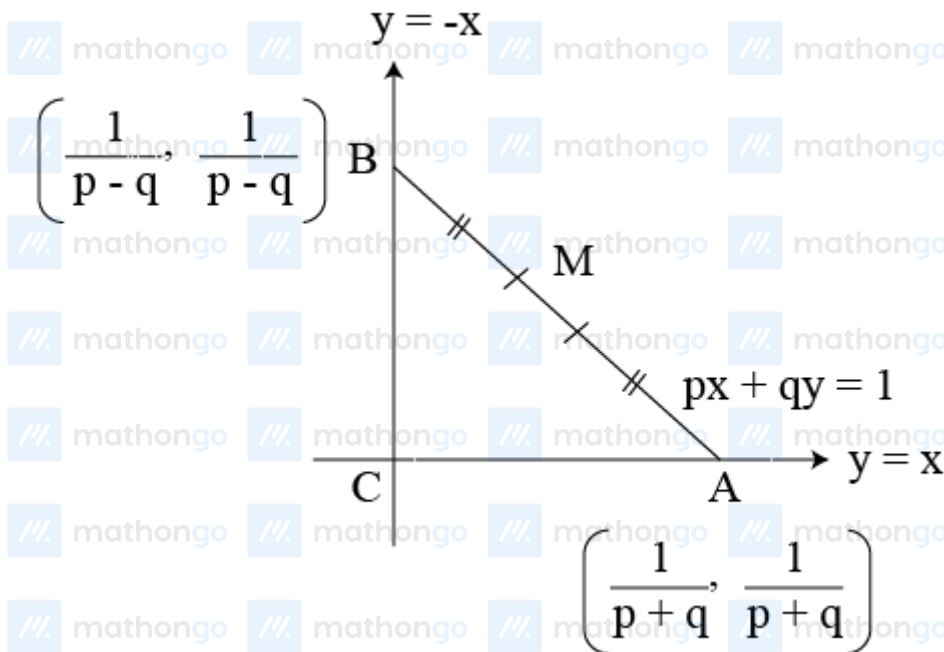
Here, A, B and C are $(-3, 1), (2, -4)$ and $(1, 3)$, respectively. $\therefore P(a, b) = \left(\frac{-3+2+1}{3}, \frac{1-4+3}{3} \right) = (0, 0)$

$$\Rightarrow a = b = 0 \dots (i)$$

$$\text{Also, } \delta = \text{ar}(\Delta ABC) = \frac{1}{2} \begin{vmatrix} -3 & 1 & 1 \\ 2 & -4 & 1 \\ 1 & 3 & 1 \end{vmatrix} = 15 \dots (ii)$$

$$\text{Thus, } a + b + \delta - 10 = 0 + 0 + 15 - 10 = 5$$

Q46 (0) - Integer Type



$$M \left(\frac{\frac{1}{p-q} + \frac{1}{p+q}}{2}, \frac{1}{p+q} - \frac{1}{p-q} \right) = \left(\frac{p}{p^2 - q^2}, \frac{-q}{p^2 - q^2} \right)$$

$$\therefore x = \frac{p}{p^2 - q^2}, y = \frac{-q}{p^2 - q^2}$$

$$\Rightarrow x^2 + y^2 = \frac{p^2 + q^2}{(p^2 - q^2)^2}$$

Hints & Solutions

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Since, $p^2 + q^2 = 1$

$$\therefore x^2 + y^2 = \frac{1}{(p^2 - q^2)^2} \quad \text{and} \quad x^2 - y^2 = \frac{1}{(p^2 - q^2)}$$

Given, $x^2 + y^2 = k(x^2 - y^2)^2$

$$\Rightarrow \frac{1}{(p^2 - q^2)^2} = k \cdot \frac{1}{(p^2 - q^2)^2}$$

$$\therefore k = 1 \Rightarrow k - 1 = 0$$

Q47 (7) - Integer Type

We have,

$$\begin{aligned} OA_n &= n \cdot A_{n-1} = n(n-1)OA_{n-1} \dots \\ &= n(n-1) \dots 2(OA_1) = n!OA_1 = n! \quad [\because OA_1 = 1] \end{aligned}$$

$$\Rightarrow \sqrt{2}(2520\sqrt{2}) = n! \Rightarrow 2520 \times 2 = n!$$

$$\Rightarrow 7 \times 6 \times 5 \times 4 \times 3 \times 2 = n!$$

$$\therefore n = 7$$

Q48 (3) - Integer Type

The sides of the triangle are given by

$$(x^2 - y^2)(2x + 3y - 6) = 0$$

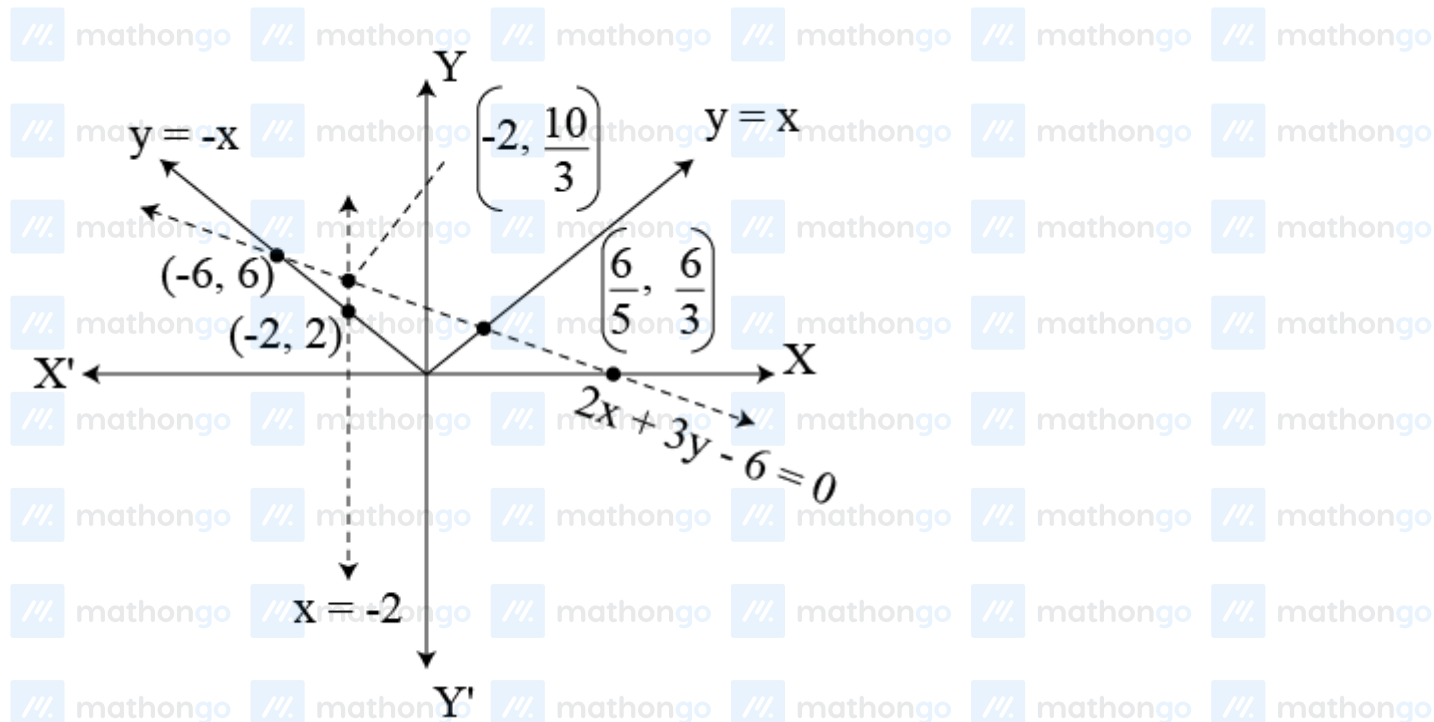
$$\text{Or } x - y = 0$$

$$x + y = 0$$

$$2x + 3y - 6 = 0$$

Hints & Solutions

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We observe that the point $(-2, a)$ moves on $x = -2$.

Thus, $2 \leq a \leq \frac{10}{3}$

$\therefore a = 3$

Q49 (12) - Integer Type

The combined equation of the straight lines joining the origin to the point of intersection of C_1 and C_2 is a

homogeneous equation and is given by

$$\text{or } \lambda x^2 + 3y^2 - 2\lambda xy + 3(3x^2 - 4y^2 + 8xy) = 0$$

$$(\lambda + 9)x^2 + 2xy(12 - \lambda) - 9y^2 = 0$$

Lines given by this equation are equally inclined with X -axis

$\therefore$ Sum of their slopes = 0

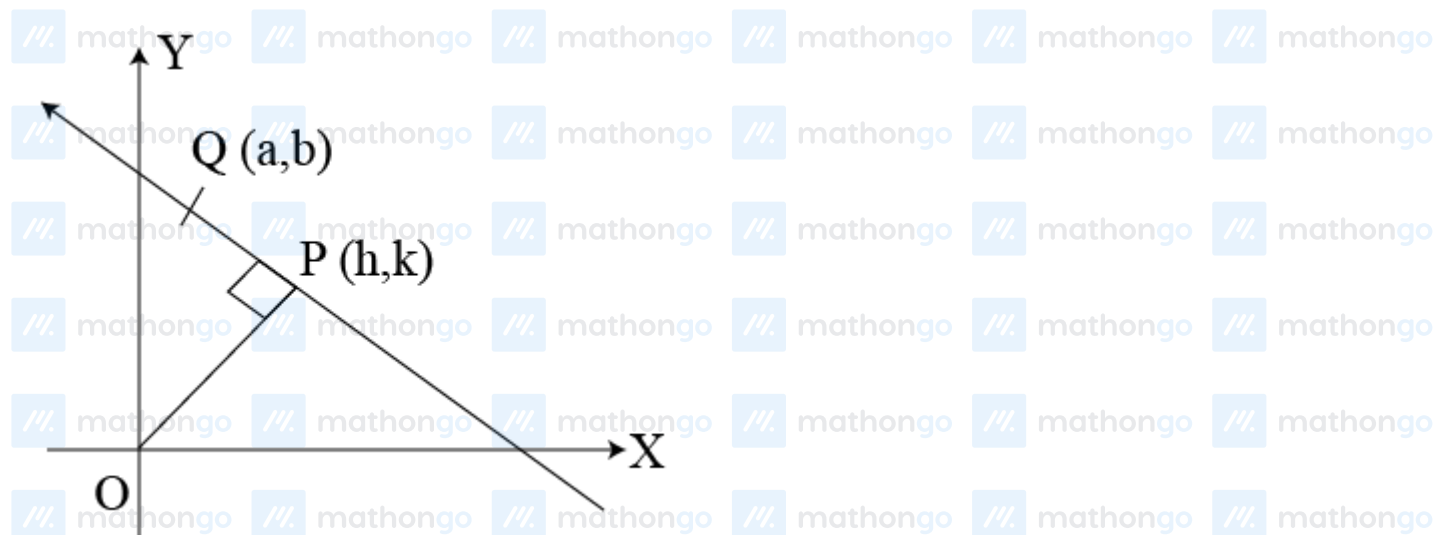
$$\Rightarrow \frac{-2(12-\lambda)}{-9} = 0 \quad \left(\because m_1 + m_2 = \frac{-2h}{b} \right)$$

$\therefore \lambda = 12$

Q50 (0) - Integer Type

Hints & Solutions

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$\therefore$ Locus of P , is $\frac{k}{h} \cdot \left(\frac{k-b}{h-a} \right) = -1$

$\Rightarrow x(x-a) + y(y-b) = 0$, locus of P

$\Rightarrow x^2 + y^2 - ax - by = 0$

$\therefore \lambda = 0$